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(a)	A photon is a quantum (i.e. a discrete amount) of energy of electromagnetic radiation.
	and its energy E , is given by $E = hf$ where h is the Planck's constant and f the frequency of the radiation.
(b)(i)	Discrete bright coloured lines on a dark background.
(b)(ii)	Each line in the emission spectrum corresponds to a specific frequency f , of the photons emitted.
	The photons are emitted when electrons transit from a higher to a lower electron energy level. Hence each energy level must be discrete.
(c)(i)	An atom is said to be in its ground state when the electrons in the atom occupies all the lowest energy states available.

(c)(ii)	Ionisation energy $E = -\frac{13.6}{\infty^2} - \left(-\frac{13.6}{1^2}\right)$ $= 13.6 \text{ eV}$ $= 13.6(1.60 \times 10^{-19})$
	$= 2.176 \times 10^{-18}$ $= 2.18 \times 10^{-18} \text{ J (3 s.f.)}$
(d)(i)	$\Delta E = \left(-\frac{2.176 \times 10^{-18}}{m^2}\right) - \left(-\frac{2.176 \times 10^{-18}}{2^2}\right)$ $\frac{hc}{\lambda} = 2.176 \times 10^{-18} \left(\frac{1}{2^2} - \frac{1}{m^2}\right)$
	$\frac{1}{\lambda} = \frac{2.176 \times 10^{-18}}{hc} \left(\frac{1}{2^2} - \frac{1}{m^2}\right)$ $R = \frac{2.176 \times 10^{-18}}{(6.63 \times 10^{-34})(3.00 \times 10^8)}$
	$= 1.094 \times 10^7$ $= 1.09 \times 10^7 \text{ m}^{-1} \text{ (3 s.f.)}$
(d)(ii)	Longest wavelength corresponds to $m = 3$ $\frac{1}{\lambda} = 1.09 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{3^2}\right)$ $= 1.514 \times 10^6$
	$\lambda = \frac{1}{1.514 \times 10^6}$ $= 6.61 \times 10^{-7} \text{ m (3 s.f.)}$
(d)(iii)	Visible (red) light
(e)(i)	Electron diffraction / (Compton scattering).
(e)(ii)	$2\pi r = 3\lambda_e$
	$\lambda_e = \frac{h}{m_e v}$
	$2\pi r = 3 \left(\frac{h}{m_e v}\right)$ $v = \frac{3h}{2\pi m_e r}$ $= \frac{3(6.63 \times 10^{-34})}{2\pi(4.78 \times 10^{-10})(9.11 \times 10^{-31})}$ $= 7.27 \times 10^5 \text{ m s}^{-1} \text{ (3 s.f.)}$

(e)(iii)	$\Delta x \Delta p \geq h$ $\Delta v \geq \frac{h}{m_e \Delta x}$ $\geq \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(0.0100 \times 10^{-9})}$
	$\geq 7.278 \times 10^7 \text{ m s}^{-1}$ $\geq 7.28 \times 10^7 \text{ m s}^{-1} \text{ (round up to 3 s.f.)}$